Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Single-Site Catalyst Platforms for Hydrogen Production via Anion Exchange Membrane Electrolysis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	Chemical Processes	Catalyst Application Technologies
Concept of Proposed Technology	This project develops single-site electrocatalyst platforms for hydrogen production via anion exchange membrane (AEM) electrolysis. The core concept is to create catalysts where the active centers are atomically dispersed or molecularly defined ("single-site") and immobilized on conductive supports, enabling high atom utilization, well-controlled catalytic environments, and improved operational stability under alkaline/AEM conditions. Compared to conventional nanoparticle catalysts (e.g., Pt/C),[1] single-site catalysts can achieve comparable catalytic function with substantially reduced precious-metal intensity by maximizing the fraction of metal atoms that are catalytically accessible. At the same time, relative to alloy catalysts commonly used in alkaline systems (e.g., Ni-based alloys),[2–3] single-site catalysts offer improved durability by minimizing restructuring and compositional drift, and by enabling rational tuning of the local coordination and interfacial chemistry that govern hydrogen evolution reaction (HER) activity and stability. The platform is designed to be flexible, allowing the choice of metal centers (e.g., earth-abundant and/or minimal precious metals), supports, and electrode configurations. The work will integrate catalyst development and screening with device-relevant evaluation and optimization. The technology targets AEM electrolyzers for green hydrogen production, where cost, durability, and scalable manufacturing are critical.			

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Rationale for Joint R&D Project	AEM water electrolysis is a promising pathway for cost-effective hydrogen production, but broader deployment is constrained by cathode catalysts and device-level durability, especially under high current density and dynamic operating conditions. While many HER catalysts show strong activity in laboratory screening, their performance and stability often change upon integration into practical electrodes and membrane–electrode assemblies (MEAs) due to ionomer/membrane interactions, mass transport, and operating-condition sensitivities. Closing this lab-to-device gap requires coordinated innovation across both catalyst design and electrolyzer integration. This joint R&D project is therefore needed to accelerate translation of single-site catalyst platforms into AEM-relevant operation. The academic team will contribute catalyst concept development, synthesis, and structure–property understanding, enabling rapid iteration and down-selection. The SME partner will provide essential capabilities in electrode/MEA fabrication, device testing, and validation under realistic conditions. Collaboration enables co-optimization of catalyst loading, electrode architecture, and operating protocols, reducing technical risk and accelerating development toward commercially viable AEM hydrogen production.
Key Words	1. Single-Site Electrocatalysts 2. Anion Exchange Membrane Water Electrolysis 3. Hydrogen Production 4. High Atom Efficiency 5. Device-Level Catalyst Integration

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	The objective of this project is to develop and validate single-site catalyst technologies for hydrogen production via anion exchange membrane (AEM) water electrolysis, advancing from proof-of-concept (TRL 4) to validation in relevant device environments (TRL 6). The project will establish the feasibility of single-site catalysts for alkaline/AEM hydrogen evolution, identify promising catalyst systems, and demonstrate their performance and stability when integrated into device-compatible electrodes and AEM electrolyzer configurations.			
R&D Target Specifications	By the end of the project, the following quantitative performance milestones will be targeted: • **Intrinsic HER activity (screening stage, alkaline electrolyte):** Overpotential ≤150 mV at 10 mA cm^{-2} in alkaline conditions, demonstrating competitive intrinsic activity for single-site catalysts. • **High-current operation (device-relevant electrodes):** Achieve ≥0.5 A cm^{-2} at practical overpotentials in alkaline or AEM-relevant configurations. • **Operational stability:** Stable hydrogen production for ≥100 hours with <10% performance loss under continuous or representative operating conditions. • **Critical-metal utilization:** Demonstrate significantly reduced precious-metal loading compared to Pt/C benchmarks (e.g., order-of-magnitude lower Pt usage where applicable) while maintaining competitive performance. • **Technology readiness:** Successful validation of selected catalysts at single-cell AEM electrolyzer level in collaboration with an SME partner. Exact benchmarking conditions, durability protocols, and comparison baselines will be finalized jointly with the SME partner to ensure industrial relevance.			
R&D Plan	The R&D plan is structured over three years, progressing from catalyst concept validation (TRL 4) to device-relevant demonstration (TRL 6) through close collaboration between the academic team and an SME partner. **Year 1: Catalyst Platform Development and Screening (TRL 4)** The first year will focus on establishing the feasibility of single-site catalyst platforms for hydrogen evolution under alkaline conditions. Candidate catalysts will be synthesized and immobilized on conductive supports using scalable and SME-compatible methods. Electrochemical screening will be carried out in liquid alkaline electrolytes to evaluate intrinsic HER activity, stability trends, and structure–performance relationships. Key outcomes include identification of promising catalyst classes, preliminary durability assessment, and down-selection of candidates for electrode integration. **Year 2: Electrode Integration and Optimization (TRL 5)** In the second year, selected catalysts will be incorporated into device-compatible electrode architectures. Efforts will focus on optimizing catalyst loading, dispersion, electrode porosity, and ionomer compatibility, recognizing that these parameters strongly influence performance in AEM systems. Testing will be conducted under higher current densities and more realistic operating conditions, including extended operation. Feedback from the SME partner will guide electrode design choices aligned with practical fabrication and integration constraints. **Year 3: Device-Level Validation and Co-Optimization (TRL 6)** The final year will emphasize AEM-relevant validation of optimized catalyst–electrode systems. In collaboration with the SME partner, electrodes will be evaluated in AEM electrolyzer under representative operating conditions. Performance metrics will include hydrogen production efficiency, current density, and operational stability. Co-optimization of catalyst properties, electrode architecture, and operating parameters will be performed to demonstrate readiness for further scale-up and commercialization.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (University of Toronto) will lead the scientific and technical foundation of the project, focusing on the development and understanding of single-site catalyst platforms for hydrogen evolution under alkaline/AEM-relevant conditions. Key responsibilities include conceptual design and synthesis of candidate catalysts, development of heterogenization strategies on conductive supports, and systematic electrochemical screening to evaluate intrinsic activity, stability, and structure–performance relationships. The institute will also be responsible for materials characterization and mechanistic analysis to support catalyst down-selection and guide iterative improvement. In addition, the research team will contribute to electrode-level integration by preparing catalyst inks, evaluating catalyst loading effects, and performing preliminary device-relevant testing in alkaline or AEM-like environments. Throughout the project, the institute will provide technical leadership, data analysis, and regular knowledge transfer to the SME partner, ensuring that catalyst development remains aligned with device integration requirements.
Expected R&R of Korean SME Partner	The Korean SME partner will lead the device integration, testing, and validation aspects of the project, contributing essential expertise in electrode fabrication, MEA preparation, and AEM electrolyzer operation. The SME will integrate selected catalysts provided by the research institute into practical electrode and electrolyzer configurations compatible with its technology platform. Key responsibilities include device-level performance evaluation, durability testing under representative operating conditions, and feedback on fabrication constraints, operating windows, and scalability considerations. The SME partner will also support co-optimization of catalyst loading, electrode architecture, and operating parameters, helping translate laboratory-scale catalyst advances into device-relevant performance. Through close collaboration, joint meetings, and data exchange, the SME will contribute to de-risking the technology and defining pathways toward scale-up and commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**This joint R&D project is expected to deliver validated single-site catalyst platforms for hydrogen production via AEM water electrolysis**, advancing the technology to TRL 5 through demonstration in device-relevant environments. Key technical outcomes include the identification of catalyst systems with high atomic utilization, competitive hydrogen evolution performance, and improved operational stability relative to conventional alloy nanoparticle catalysts under alkaline/AEM conditions. From a commercialization perspective, the project will generate de-risked catalyst and electrode concepts that are compatible with SME-led electrolyzer manufacturing and integration workflows. **By reducing reliance on high precious-metal loadings while maintaining device-level performance, the technology has the potential to lower system cost and enhance durability, two critical barriers to wider adoption of AEM electrolyzers.** The collaborative framework will also establish a foundation for follow-on development, pilot-scale testing, and industrial deployment with the SME partner. More broadly, the project will contribute to the advancement of cost-effective and scalable green hydrogen technologies, supporting decarbonization of energy and industrial sectors. The outcomes will strengthen international collaboration, build technical capacity in catalyst–device co-development, and enhance the competitiveness of participating institutions and industrial partners in the global hydrogen economy.

6. References

(1) *J. Am. Chem. Soc.* 2015, 137, 13, 4347–4357

(2) *ACS Catal.* 2023, 13, 17, 11589–11597

(3) *Adv. Energy Mater.* 2025, 15, 2501800

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization